

Submission Summary

Conference Name

7th International Conference on Data Science and Applications

Paper ID

2564

Paper Title

Improving Chatbot Capabilities to Enhance Conversational AI

Abstract

Modern conversational AI systems may encounter several difficulties when understanding users' intentions, keeping track of their context during the multi-turn process, and providing relevant replies. The traditional rule-based chatbots are inflexible and do not account for dynamic user input, which negatively affects their performance and limits the possibilities for their scaling. To address these problems, this research project proposes implementing NLP strategies into chatbots to improve their performance. The experimental design includes creating two chatbot systems: a conventional one using the rule-based algorithm and a novel solution relying on modern machine learning approaches, such as transformer-based algorithms (BERT and GPT). The two models will be assessed through different conversational scenarios: ambiguous queries, sentiment-based interaction, and multi-turn conversations. Both models will be evaluated for response accuracy, context awareness, and users' satisfaction. The results show that the novel chatbot system outperforms the traditional solution, providing more accurate replies and a better understanding of user intentions and context.

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Primary Subject Area

Signal & Image Processing

Secondary Subject Areas

Data Science

Submission Files

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Submission Questions Response

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